

Effect of Drug X on Outcome Y: a randomised controlled trial

1. Title and abstract

Effect of Drug X on Outcome Y: a randomised controlled trial.

Background: Outcome Y remains poorly controlled with standard care. **Methods:** We conducted a two-arm, parallel-group randomised trial. **Results:** The mean difference in outcome Y was 4.2 points in favour of Drug X. **Conclusions:** Drug X improves outcome Y relative to placebo.

2. Open science

Registered at ClinicalTrials.gov, NCT01234567, registered 2024-01-15.

The protocol and statistical analysis plan are available at osf.io/example.

This trial was funded by the National Research Agency (grant 42). The funder had no role in the design, conduct, or reporting of the trial.

There is no item 5c in CONSORT 2025 — this call exists only to exercise the “id inconnu” diagnostic.

3. Introduction

Prior trials suggested a plausible mechanism of action for Drug X on outcome Y.

The primary objective was to determine whether Drug X reduces outcome Y compared with placebo, and to characterise its harms.

4. Methods

Adults aged 18-75 with a confirmed diagnosis of condition Z.

Participating sites were university hospitals with a dedicated research nurse.

Participants received either Drug X (10 mg once daily) or a matching placebo for 12 weeks.

The primary outcome was change in score S from baseline to week 12, analysed as a mean difference between groups.

Harms were monitored via routine adverse-event reporting at each study visit.

The sample size of 240 participants was calculated to detect a 4-point difference in score S with 90% power.

4.1. Randomisation

The allocation sequence was generated by an independent statistician using a computer-based random-number generator.

Randomisation used permuted blocks of 4 and 6, stratified by site.

Allocation was concealed using sequentially numbered, opaque, sealed envelopes prepared by a party independent of recruitment.

Research nurses enrolling participants had no access to the allocation sequence until after enrolment.

Participants and outcome assessors were blinded to group assignment; care providers were not.

Groups were compared for the primary outcome using a mixed-effects linear model.

All randomised participants were analysed in their assigned group, regardless of adherence (intention-to-treat).

Missing outcome data were handled by multiple imputation under a missing-at-random assumption.

5. Results

As stated in the Methods, outcome S was assessed by a rater blinded to group assignment. 240 participants were randomly assigned: 120 to each group. Recruitment ran from March 2024 to September 2024, with follow-up completed in December 2024. Table 1 shows baseline characteristics for each group.

Characteristic	Drug X	Placebo
Age, mean (SD)	52 (12)	54 (11)
Female, n (%)	61 (51%)	58 (48%)

Table 1: Baseline characteristics.

For the primary outcome, the mean difference between groups was 4.2 points (95% CI 1.1 to 7.3). No serious adverse events were reported in either group.

6. Discussion

These results are consistent with the hypothesised mechanism of action of Drug X, and are broadly in line with prior smaller trials. This trial was limited by its single-country setting and its 12-week follow-up, which may not capture longer-term harms.